MISC

Quadratic formula:

$$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Monotonically in-/decreasing:

$$\forall x.\ (f'(x) > 0) \Rightarrow a > b \Leftrightarrow f(a) > f(b)$$
$$\forall x.\ (f'(x) < 0) \Rightarrow a > b \Leftrightarrow f(a) < f(b)$$

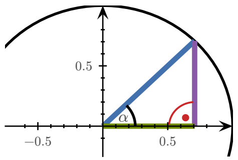
RSS:

$$\mathrm{RSS} = \sum_i (y_i - f(x_i))^2$$

TRIGONOMETRY

x	0	$30° = {}^{\pi}/_{6}$	$45° = {}^{\pi}/_{4}$	$60° = {}^{\pi}/_{3}$	$90° = {}^{\pi}/_{2}$
$\sin(x)$	0	${}^{1}/_{2}$	${}^{\sqrt{2}}/_{2}$	${}^{\sqrt{3}}/_{2}$	1
$\cos(x)$	1	${}^{\sqrt{3}}/_{2}$	${}^{\sqrt{2}}/_{2}$	${}^{1}/_{2}$	0
$\tan(x)$	0	${}^{1}/_{\sqrt{3}}$	1	$\sqrt{3}$	

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$
$$\sin(2x) = 2\sin(x)\cos(x)$$
$$\sin^2(x) + \cos^2(x) = 1$$
$$1, \sin(\alpha), \cos(\alpha)$$



VECTORS

Term	Definition
Norm/Length	$\lVert v\rVert = \sqrt{\sum_{i=1}^n v_i^2} = \sqrt{v^\top v}$ $\nabla\lVert v\rVert = \frac{1}{\lVert v\rVert} v^\top$ $\nabla\frac{1}{\lVert v\rVert} = -\frac{1}{\lVert v\rVert^2}\nabla\lVert v\rVert = -\frac{1}{\lVert v\rVert^3}v^\top$ $\nabla\lVert r\rVert^2 = 2\lVert r\rVert\frac{1}{\lVert r\rVert}r^\top \quad = 2r^\top$
Distance	$d(u,v) = \lVert u - v\rVert$ $\nabla d(u,v) = \frac{1}{d(u,v)}((u-v)^\top (v-u)^\top)$
Scalar product	$u \bullet v = u^\top v = \sum_k u_k v_k$ $\nabla(u \bullet v) = (v^\top\ u^\top)$ $\nabla_u(u \bullet v) = v^\top$

MATRICES

Affine transformation:

$$M \in \mathbb{R}^{m\times n}, \quad A_{b,M} : \begin{cases} \mathbb{R}^m \to \mathbb{R}^n \\ x \quad \mapsto b + M\cdot x \end{cases}$$

Determinant:

$$\det(a) = a \qquad \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

$$\det(A\cdot B) = \det(A)\cdot\det(B) \qquad \det(\mathbb{1}) = 1$$

$$\det(A^{-1}) = \frac{1}{\det(A)} \qquad \det(A^\top) = \det(A)$$

Inverting:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Transposing:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^\top = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix} \quad (A^\top)^\top = A \quad (A+B)^\top = A^\top + B^\top$$

$$\begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^\top = (1\ 4\ 5) \quad (\lambda A)^\top = \lambda A^\top \quad (AB)^\top = B^\top A^\top$$

Matrix multiplication:

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix}$$
$$= \begin{pmatrix} (2\cdot 6 + 3\cdot 1 + 1\cdot 8) & (2\cdot 4 + 3\cdot 0 + 1\cdot 9) \\ (4\cdot 6 + -1\cdot 1 + 7\cdot 8) & (4\cdot 4 + -1\cdot 0 + 7\cdot 9) \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$$

PROPERTIES

$$A\cdot A^{-1} = \mathbb{1} \qquad \ker A = \{v \mid Av = 0\}$$
$$A + B = B + A \qquad A\cdot B \neq B\cdot A$$
$$A + (B + C) = (A + B) + C \qquad A(BC) = (AB)C$$
$$A(B + C) = AB + AC \qquad (A + B)C = AC + BC$$

INDICES

$$A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{m\times r}, \qquad A\cdot B \in \mathbb{R}^{n\times r}$$
$$(A\cdot B)_{ij} = \sum_{k=1}^m A_{ik}B_{kj} \qquad = \sum_k A_{ik}B_{kj}$$
$$((A\cdot B)^\top)_{ij} = (A\cdot B)_{ji} \qquad = \sum_k A_{jk}B_{ki}$$
$$(A\cdot B^\top)_{ij} = \sum_k A_{ik}B_{kj}^\top \qquad = \sum_k A_{ik}B_{jk}$$
$$a^\top \cdot B \cdot b = \sum_{ij} (a^\top)_i B_{ij} b_j \qquad = \sum_{ij} a_i B_{ij} b_j$$
$$(B\cdot a)(C\cdot a) = \sum_i (B\cdot a)_i (C\cdot a)_i$$

ADDITION/SUBTRACTION

$$A \in \mathbb{R}^{n\times m}, B \in \mathbb{R}^{n\times m}$$
$$(A + B)_{ij} = A_{ij} + B_{ij}$$
$$(A - B)_{ij} = A_{ij} - B_{ij}$$

KRONECKER DELTA

$$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1 & , i = j \\ 0 & , \text{otherwise} \end{cases}$$
$$(E_{ij})_{kr} = \delta_{ik}\delta_{jr}$$
$$(E_{rst})_{ijk} = \delta_{ri}\delta_{sj}\delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x) = a + M\cdot x$$
$$L_A(\vec{0}) = \vec{0}$$

DERIVATIVES

1	$\to 0$	x	$\to 1$
x^2	$\to 2x$	x^n	$\to nx^{n-1}$
$\frac{1}{x}$	$\to -\frac{1}{x^2}$	$\sqrt{x}$	$\to \frac{1}{2\sqrt{x}}$
e^x	$\to e^x$	e^{-x}	$\to -e^{-x}$
$\ln(x)$	$\to \frac{1}{x}, x > 0$	$\ln(y\cdot x)$	$\to \frac{1}{x}, x > 0$
a^x	$\to \ln(a)\cdot a^x, a > 0, a \neq 1$	$\log_b(x)$	$\to \frac{1}{\ln(b)\cdot x}$
$\sin(x)$	$\to \cos(x)$		
$\cos(x)$	$\to -\sin(x)$	$\sin(2x)$	$\to 2\cos(2x)$
$\tan(x)$	$\to 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\cos(ax)$	$\to -a\sin(ax)$
$\cot(x)$	$\to -\frac{1}{\sin^2(x)}$	$\operatorname{arccot}(x)$	$\to -\frac{1}{1+x^2}$
$\arccos(x)$	$\to -\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x)$	$\to \frac{1}{\sqrt{1-x^2}}$
		$\arctan(x)$	$\to \frac{1}{1+x^2}$

Case	Rule
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha\cdot f(x))' = \alpha\cdot f'(x)$
Product rule	$(f\cdot g)' = f'\cdot g + f\cdot g'$ $(f\cdot g\cdot h)' = f'\cdot g\cdot h + f\cdot g'\cdot h + f\cdot g\cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x))\cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES

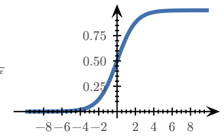
Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
n-dimensional curve	$c : \mathbb{R} \to \mathbb{R}^n$
n-dimensional hyper-surface	$s : \mathbb{R}^n \to \mathbb{R}$
Reparametrization	$c(t) \to d(s) = c(h(s))$

Softmax function:

$$\sigma : \begin{cases} \mathbb{R}^n \to [0;1]^n \\ x_i \ \mapsto e^{x_i}\left(\sum_{j=1}^n e^{x_j}\right)^{-1} \end{cases}$$
$$J_{ij}\sigma(x) = \delta_{ij}\frac{e^{x_i}}{\sum_{k=1}^n e^{x_k}} - \frac{e^{x_i}e^{x_j}}{\left(\sum_{k=1}^n e^{x_k}\right)^2}$$

Sigmoid function:

$$\sigma : \begin{cases} \mathbb{R} \to [0;1] \\ x \mapsto \frac{1}{1+e^{-x}} = \frac{e^x}{1+e^x} \end{cases}$$
$$\sigma(-x) = 1 - \sigma(x)$$
$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



DERIVATIVES

Function	Derivative
$f : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases}$	$f' : \begin{cases} \mathbb{R} \to \mathbb{R}^n \\ x \mapsto y \end{cases} = \begin{pmatrix} \frac{\mathrm{d}}{\mathrm{d}x}f_1(x) \\ \frac{\mathrm{d}}{\mathrm{d}x}f_2(x) \\ \vdots \\ \frac{\mathrm{d}}{\mathrm{d}x}f_n(x) \end{pmatrix}$
$f : \begin{cases} \mathbb{R}^n \to \mathbb{R} \\ x \quad \mapsto y \end{cases}$	$f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{1\times n} \\ x \quad \mapsto y \end{cases} = \nabla f$ $= \left(\frac{\partial}{\partial x_1}f(x), \frac{\partial}{\partial x_2}f(x), \ldots, \frac{\partial}{\partial x_n}f(x)\right)$
$f : \begin{cases} \mathbb{R}^n \to \mathbb{R}^m \\ x \quad \mapsto y \end{cases}$	$f' : \begin{cases} \mathbb{R}^n \to \mathbb{R}^{m\times n} \\ x \quad \mapsto y \end{cases} = J_f = \left.\frac{\partial f}{\partial x}\right\rvert_{x=x_0}$ $= \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1}f_1(x) & \ldots & \frac{\partial}{\partial x_n}f_1(x) \\ \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1}f_m(x) & \ldots & \frac{\partial}{\partial x_n}f_m(x) \end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE

$$f(x,y) = x^2 + 2y$$
$$\frac{\partial f}{\partial x} = 2x$$
$$\frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 2\cdot\frac{\mathrm{d}y}{\mathrm{d}x} \mid (\text{y dependent on x})$$

GRADIENT VS TOTAL DERIVATIVE

$$f(x,y) = xye^y$$
$$x(t) = e^t$$
$$y(t) = t^2$$
$$\nabla f = (ye^y, xe^y + xye^y) \qquad = e^y(y, x + xy)$$
$$\frac{\mathrm{d}}{\mathrm{d}t}f = e^y(y, x + xy)\cdot\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} e^t \\ t^2 \end{pmatrix} = \frac{\mathrm{d}}{\mathrm{d}t}\left(e^t t^2 e^{t^2}\right)$$
$$= (2t + t^2 + 2t^3)e^{t^2+t}$$

GRADIENTS

Common gradients:

$$f : \mathbb{R}^n \to \mathbb{R},\ x \in \mathbb{R}^n, \quad a \in \mathbb{R}^n,\ c \in \mathbb{R},\ A \in \mathbb{R}^{n\times n}$$

$$f(x) = \lVert x\rVert^2 \qquad \Rightarrow \nabla f = 2x^\top$$
$$f(x) = a^\top x + c \qquad \Rightarrow \nabla f = a^\top$$
$$f(x) = x^\top Ax \qquad \Rightarrow \nabla f = x^\top(A + A^\top)$$
$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c \quad \Rightarrow \nabla f = x^\top\frac{A + A^\top}{2} + b^\top$$
$$= x^\top A + b^\top \text{ if } A^\top = A$$
$$g : \mathbb{R}^m \to \mathbb{R}, B \in \mathbb{R}^{m\times n}$$
$$f(x) = g(Bx + b) \qquad \Rightarrow \nabla f = (\nabla g(Bx + b))^\top B$$

JACOBIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}^m \qquad x_0 \in \mathbb{R}^n$$
$$J_f(x_0) : \mathbb{R}^{m\times n} \qquad J_f(x_0)_{ij} = \frac{\partial}{\partial x_j}f_i(x_0)$$
$$f : \mathbb{R}^2 \to \mathbb{R}^2 \qquad J_f(x,y) = \begin{pmatrix} \partial x_1 & \partial y_1 \\ \partial x_2 & \partial y_2 \end{pmatrix}$$

ty lukas <3

$$\begin{tikzcd} \mathrm{Euclid}_* \\ (\mathbb{R}^n, x_0) \arrow[d, "g"'] \arrow[dd, bend left, "f\circ g"] \\ (\mathbb{R}^m, g(x_0)) \arrow[d, "f"'] \\ (\mathbb{R}^k, f(g(x_0))) \end{tikzcd} \overset{D}{\Rightarrow} \begin{tikzcd} \mathrm{Mat}_{\mathbb{R}} \\ \mathbb{R}^n \arrow[d, "J_g(x_0)"'] \arrow[dd, bend left, "J_f(g(x_0))J_g(x_0)"] \\ \mathbb{R}^m \arrow[d, "J_f(g(x_0))"'] \\ \mathbb{R}^k \end{tikzcd}$$

Chain rule:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R}^k,\ h : \mathbb{R}^n \to \mathbb{R}^k = g\circ f$$
$$J_h = J_g(f(x))\cdot J_f(x)$$

Common jacobians:

$$f(x) = a + M\cdot x \qquad \Rightarrow J_f = M$$
$$f(x) = x \qquad \Rightarrow J_f = \mathbb{1}$$
$$f(x) = \lVert x\rVert^2 \qquad \Rightarrow J_f = (2x)^\top$$
$$f_i : \begin{cases} \mathbb{R}^n \to \mathbb{R}^n \\ x \quad \mapsto a_i + M_i x \end{cases}$$
$$g(x) = f_n(f_{n-1}(\ldots(f_1(x)))) \Rightarrow J_g = M_n\cdot M_{n-1}\cdot\ldots\cdot M_1$$

LINEARISATION

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ x_0 \in \mathbb{R}^n$$
$$L_f(x) = f(x_0) + J_f(x_0)\cdot(x - x_0)$$

For finding better approximations for x_0:
1) Calculate J_f, then $J_f(x_0)$, then L_f
2) Find x so that $L_f(x) \overset{!}{=} 0$ (Gauss)

For calculating an approximate value at x:
1) x_0 = round x to ints
2) Calculate $L_f(x)$ at x_0

SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right) = \frac{\partial^2 f(x,y)}{\partial x^2} = \partial_{xx}f(x,y)$$

HESSIAN MATRIX

$$f : \mathbb{R}^n \to \mathbb{R}, \quad x \in \mathbb{R}^n, \quad H(x) \in \mathbb{R}^{n\times n}$$
$$H(x,y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$$
$$(H(x))_{ij} = \frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(x)$$
$$\partial_i\partial_j f = \partial_j\partial_i f$$

Common hessians:

$$f(x) = \frac{1}{2}x^\top Ax + b^\top x + c \Rightarrow H_f = \frac{A + A^\top}{2}$$
$$f(x) = g(Ax + b) \qquad \Rightarrow H_f = A^\top H_g(Ax + b)A$$

$$g : \mathbb{R}^m \to \mathbb{R}$$

EXTREMAL POINTS

Quadratic form:

$$M \in \mathbb{R}^{n\times n} \qquad q_M(v) = v^\top M v \qquad v \in \mathbb{R}^n$$

$$H = \frac{1}{2}(M + M^\top) \quad \forall v \in \mathbb{R}^n.\ (q_M(v) = q_H(v))$$

$v^\top H v > 0 \Rightarrow c_0$ is a ***local minimum***, H is ***positive definite***

$v^\top H v \geq 0 \Rightarrow c_0$ is a ***local minimum*** or ***non-extremal***, H is ***positive semi-definite***

$v^\top H v < 0 \Rightarrow c_0$ is a ***local maximum***, H is ***negative definite***

$v^\top H v \leq 0 \Rightarrow c_0$ is a ***local maximum*** or ***non-extremal***, H is ***negative semi-definite***

$v^\top H v \neq 0 \Rightarrow c_0$ is a ***saddle point***, H is ***indefinite***

Finding stationary points:

1) Calculate ∇f, then H_f
2) Find stationary point(s) x so that $\nabla f(x) \stackrel{!}{=} 0$
3) Find type of point using technique of completing squares on $H_f(x)$

$$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1v_2h_{12} + 2v_2v_3h_{23} + 2v_1v_3h_{13}$$

Technique of completing squares:

$$\underbrace{x^2}_{\sqrt{\cdot}} - \underbrace{4xy}_{\div 2x} + 1$$

$$= (x - 2y)^2 - (-2y)^2 + 1$$

$$= (x - 2y)^2 - 4y^2 + 1$$

HYPER-SURFACES

Chain rule:

$$f : \mathbb{R}^n \to \mathbb{R}^m,\ g : \mathbb{R}^m \to \mathbb{R},\ h : \mathbb{R}^n \to \mathbb{R} = g \circ f$$

$$\nabla h(x) = \nabla g(f(x)) = \nabla g(f)|_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$$

Differentiation properties:

$$f : \mathbb{R}^n \to \mathbb{R},\ g : \mathbb{R}^n \to \mathbb{R}$$

$$\nabla(f + g) = \nabla f + \nabla g$$

$$\nabla(f - g) = \nabla f - \nabla g$$

$$\nabla(f \cdot g) = g\nabla f + f\nabla g$$

$$\nabla\frac{f}{g} = \frac{g\nabla f - f\nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$f : D \to M$

$g : N \to R$

$h : D \to R$

$= g \circ f$

$$\begin{tikzcd} D \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] & M \arrow[r, "g"] & R \end{tikzcd}$$

GRADIENT DESCENT / NEWTON'S METHOD

$$GD \qquad x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$$

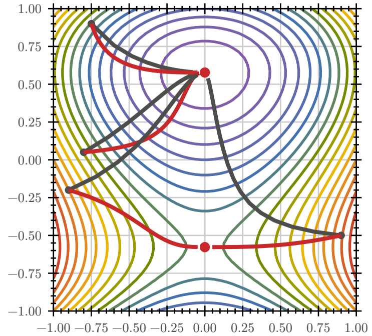
$$N \qquad x_{i+1} = x_i - \gamma \cdot \left(H_{f(x_i)}\right)^{-1} \cdot \nabla f(x_i)$$

$$\gamma = \text{step size}$$

$$\text{termination criterion } \varepsilon > |x_{i+1} - x_i|$$

Gradient descent doesn't always produce global minimum:

- If the function has more than one local minimum it could output that → run with different starting points
- Could output a stationary point that is not minimal (unlikely). → run with different starting points
- Could not terminate at all if step size / learning rate is so big (overshoots minimal point). Likely to happen if the graph of the function is narrow around the minimal point. → choose smaller step size



LOGISTIC REGRESSION

$$\begin{tikzcd} \mathbb{R}^{128\times256\times3} \arrow[r, "b(x)"] \arrow[rrd, "cat(x)"] & \mathbb{R} \arrow[r, "\sigma(m \cdot b - q)"] & {[0;1]} \arrow[d, "\mathbb{1}_{(t;\infty)}(\sigma)"] \\ & & \{0;1\} \end{tikzcd}$$

$$\begin{aligned} b(x) &= \text{Feature function} \\ \sigma(m \cdot b - q) &= \text{Probability function} \\ m &= \text{Weight} \\ q &= \text{Bias} \\ \mathbb{1}_{(t;\infty)}(\sigma) &= \text{Indicator function} \\ t &= \text{Threshold} \end{aligned}$$

TODO:
- FS2023 4)
- $l(p) = (\prod p) \cdot (\prod(1 - p))$

PROBABILITY THEORY

Term	***Definition***
Distributed	$\sim$
Normal distribution	$\mathcal{N}(\mu, \sigma)$
Uniform distribution	$\text{unif}(a, b)$
Probability measure	$\mathbb{P}(E)$
Density	$f_\theta(x_i)$
Likelihood function	$L(\theta) = \prod_{i=1}^{n} f_\theta(x_i)$
Log-likelihood function	$\ln(L(\theta)) = \sum_{i=1}^{n} \ln(f_\theta(x_i))$
Cost function (negative log-likelihood)	$c(\theta) = -\ln(L(\theta)) = -\sum_{i=1}^{n} \ln(f_\theta(x_i))$
Indicator function	$\mathbb{1}_{(a;b)}(x) = \begin{cases} 1 & x \in (a;b) \\ 0 & x \notin (a;b) \end{cases}$
CDF	Cumulative Distribution Function F
PDF	Probability Density Function f

$$|f_Y(y)\,\mathrm{d}y| = |f_X(x)\,\mathrm{d}x| \quad \text{if monotonically in/decreasing}$$

MAXIMUM LIKELIHOOD PRINCIPLE

= Finding optimal parameters θ for fitting distribution to data

1) Calculate cost function c with given distribution function f_θ, parameters θ and data points $x_1, x_2, ..., x_n$
2) Calculate stationary point(s) of c (set ∇c to 0)
   ⇒ Find optimum (gradient descent/newton's method if calculation not possible)

KOLMOGOROV AXIOMS

$$\mathbb{P}(\Omega) = 1$$

$$\mathbb{P}(\varnothing) = 0$$

$$\forall A, B \subset \Omega,\ A \cap B = \varnothing.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$$

$$\forall A, B \subset \Omega.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$$

STATISTICALLY INDEPENDENT

$X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ are ***statistically independent***, if the probability density f_V of the random variable

$$V : \begin{cases} \Omega \to \mathbb{R}^2 \\ \omega \mapsto \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} \end{cases}$$

satisfies

$$f_V(x, y) = f_X(x) \cdot f_Y(y)$$

DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega = \{e_1, e_2, ..., e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, ..., n\}, \quad E \subset \Omega$$

$$\mathbb{P}(E) = \sum_{e \in E} p(e)$$

$$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$$

RULES

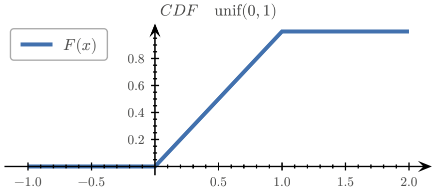
$$\forall e \in \Omega.\ (0 \leq p(e) \leq 1)$$

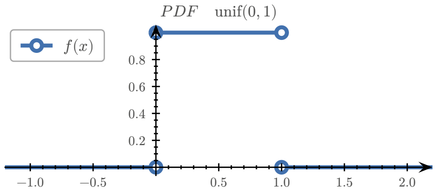
$$\sum_{e \in \Omega} p(e) = 1 \qquad \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t = 1$$

$$\lim_{x \to -\infty} F(x) = 0 \qquad \lim_{x \to \infty} F(x) = 1$$

$$0 \leq F(x) \leq 1 \qquad f(t) \geq 0 \text{ for } t \in \mathbb{R}$$

$F(x)$ is monotonically increasing & continuous from the right





UNIVARIATE UNIFORM DISTRIBUTION

$$X \sim \text{unif}(a, b), \quad \Omega \subset \mathbb{R}$$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a;b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) \cdot \frac{1}{b - a} = F_X'$$

$$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) \quad = \mathbb{1}_{[a;b]}(x) \cdot \frac{x - a}{b - a} + \mathbb{1}_{(b;\infty)}(x)$$

$$= \mathbb{P}(X \leq x) \quad = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$$

UNIVARIATE NORMAL DISTRIBUTION

Standard normal distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

General normal distribution:

$$f(x|\mu, \sigma) = \frac{1}{\sigma}\varphi\left(\frac{x - \mu}{\sigma}\right)$$

Univariate normal distribution:

$$x \sim \mathcal{N}(\mu, \sigma)$$

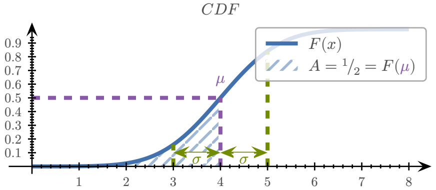
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

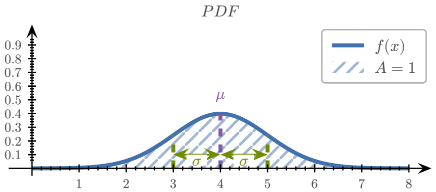
$$\mu \approx \overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$\sigma^2 \approx \text{var} = \frac{1}{N - 1}\sum_{i=1}^{N} (x_i - \overline{x})^2$$

Error function:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$$





mean value (μ), standard deviation (σ)

MULTIVARIATE NORMAL DISTRIBUTION

$$\Sigma = \text{covariance matrix}, \quad V : \Omega \to \mathbb{R}^n$$

$$V \sim \mathcal{N}(\mu, \Sigma)$$

$$f_V(v) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \cdot e^{q_{\Sigma^{-1}}(v - \mu)}$$

$$q_{\Sigma^{-1}}(v - \mu) = -\frac{1}{2}(v - \mu)^\top \Sigma^{-1}(v - \mu)$$

Sample mean:

$$\mu \approx \overline{v} = \frac{1}{N}\sum_{i=1}^{N} v_i$$

Sample covariance matrix:

$$\Sigma \approx Q = \frac{1}{N - 1}\sum_{i=1}^{N} (v_i - \overline{v})(v_i - \overline{v})^\top$$

EXAMPLES

WORKING WITH INDICES

$$x, y \in \mathbb{R}^n, \quad f : \mathbb{R}^n \to \mathbb{R}$$

$$f(x) = (x - y)^\top(3x - y) = \sum_{i=1}^{n} (x_i - y_i)(3x_i - y_i)$$

$$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$$

$$f(x) = \ln(x_1 \cdot x_2 \cdot ... \cdot x_n) - |x|^2 = \sum_{i=1}^{n} (\ln(x_i) - x_i^2)$$

$$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$$

$$f(x) = \ln(y^\top x) = \ln\left(\sum_{i=1}^{n} y_i x_i\right) \Rightarrow \nabla f = \frac{1}{y^\top x} y^\top$$

CALCULATING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

x	y
1	7
2	16
3	32
4	44

$$f(t) = \lambda e^{-\lambda t} \cdot 1_{(0;\infty)}(t)$$

find a formula for the likelihood function:

$$l(\lambda) = f(7) \cdot f(16) \cdot f(32) \cdot f(44) = \lambda^4 e^{-99\lambda}$$

Find a formula for the negative log-likelihood:

$$-\ln(l(\lambda)) = -\ln(\lambda^4 e^{-99\lambda}) = 99\lambda - 4\ln(\lambda)$$

CALCULATING PROBABILITY DENSITY FUNCTION

EXPONENTIAL DISTRIBUTION

Given σ, Y and X, an exponentially distributed random var

$$f_X(x) = X_{[0;\infty)}(x) \cdot e^{-x}$$

$$Y = \sigma(X)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

calculate the probability density function $f_Y(y)$

$$|f_Y(y)\,\mathrm{d}y| = |f_X(x)\,\mathrm{d}x|$$

$$\stackrel{\frac{\mathrm{d}x}{\mathrm{d}y} > 0}{\Leftrightarrow} f_Y(y) = f_X(x) \cdot \frac{\mathrm{d}x}{\mathrm{d}y}$$

$$y = \frac{1}{1 + e^{-x}}$$

$$\Leftrightarrow x = -\ln\left(\frac{1}{y} - 1\right), \quad 0 < y < 1$$

$$\Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{1}{\frac{1}{y} - 1} \cdot \frac{-1}{y^2} = \frac{1}{y - y^2}$$

$$\Rightarrow f_Y(y) = X_{[0;\infty)}(x) \cdot e^{-x} \cdot \frac{1}{y - y^2}$$

$$= \frac{1}{y^2} \cdot X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right)$$

$$-\ln\left(\frac{1}{y} - 1\right) \geq 0$$

$$\stackrel{y > 0}{\Leftrightarrow} y \geq \frac{1}{2}$$

$$\Rightarrow X_{[0;\infty)}\left(-\ln\left(\frac{1}{y} - 1\right)\right) \stackrel{y \in (0;1)}{=} X_{[\frac{1}{2};1)}(y)$$

$$\Rightarrow f_Y(y) = \frac{1}{y^2} X_{[\frac{1}{2};1)}(y)$$

UNIFORM DISTRIBUTION

Given $X \sim \text{unif}(0, 1)$, $p > 0$, $Y = X^p$ calculate the probability density function $f_Y(y)$

$$F_X(x) = \mathbb{P}(X \leq x)$$

$$F_Y(y) = \mathbb{P}(Y \leq y) \qquad = \mathbb{P}(X^p \leq y)$$

$$= \mathbb{P}\left(X \leq y^{\frac{1}{p}}\right) \qquad = F_X\left(y^{\frac{1}{p}}\right)$$

$$= \mathbb{1}_{[0;1]}(y) y^{\frac{1}{p}} + \mathbb{1}_{1;\infty}(y)$$

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} F_Y(y) = \mathbb{1}_{[0;1]}(y) \frac{1}{p} y^{\frac{1}{p} - 1}$$

UNIFORM DISTRIBUTION 2

$$X \sim \text{unif}(1, 3), \quad Y = \sqrt[3]{\frac{5 - X}{2}}$$

$$f_X(x) = \mathbb{1}_{(1;3)}(x)\frac{1}{2}$$

$$X = 5 - 2Y^3$$

$$g(y) = 5 - 2y^3$$

$$f_Y(y) = -f_X(g(y))g'(y) \qquad \text{monot. decr.}$$

$$= 3y^2 \mathbb{1}_{(1;3)}(5 - 2y^2)$$

$$1 < 5 - 2y^2 < 3$$

$$\Leftrightarrow 1 < y < \sqrt[3]{2}$$

$$\Rightarrow f_Y(y) = 3y^2 \mathbb{1}_{(1;\sqrt[3]{2})}(y)$$

TODO:
- check if function is increasing ($f' > 0$) → Aufgabe 123/124
- distribution to density and vice-versa $X = g(Y) \wedge g$ increasing $\Rightarrow f_Y(y) = f_X(g(y)) \cdot g'(y)$

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\hat{\sigma} = \left(\frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2 \right)^{\frac{1}{2}}$$